



North South University
Department of Electrical & Computer Engineering

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Course Title: Analog Electronics I Lab

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Section: 7

Experiment Number:01

Experiment Name:

I-V Characteristics of diode

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Experiment No: 1

I-V characteristics of diode

Objectives:

- ☐ To study and analyze the current-voltage (I-V) characteristics of a p-n junction diode under both forward and reverse bias conditions, and to
- ☐ To determine its threshold voltage and behavior in different biasing regions.

Theory:

Diode is a two terminal device. A diode ideally conducts in only one direction.

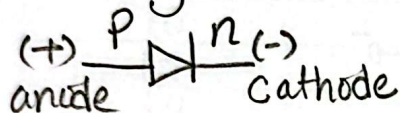


Fig: Schematic Diagram of diode

Diode has two characteristics. They are:-

Conduction region: For ideal diode, The voltage across the diode is 0V and the diode acts like a short circuit.

Non-conduction region: For ideal diode, All the voltage is across the diode. The current is 0A. The diode acts like open.

Diode has three operating conditions. They are:-

No bias: For ideal diode, No extra voltage is applied, $V_d = 0V$ and there is no diode current, $I_d = 0A$.

Forward bias: For ideal diode, external voltage is applied in the p-n with the same polarity as the p and n-type materials.

Reverse bias: For ideal diode, external voltage is applied ~~across~~ across the p-n junction with the opposite polarity of the p and n-type materials. Here diode current is 0A and diode voltage $V_d = E$

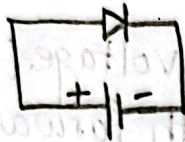


Fig: Forward bias

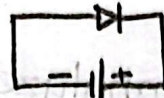


Fig: Reverse bias

Static resistance: For a specific applied DC voltage (V_d), the diode has a specific current (I_d) and a specific resistance (R_d) where, $R_d = \frac{V_d}{I_d}$

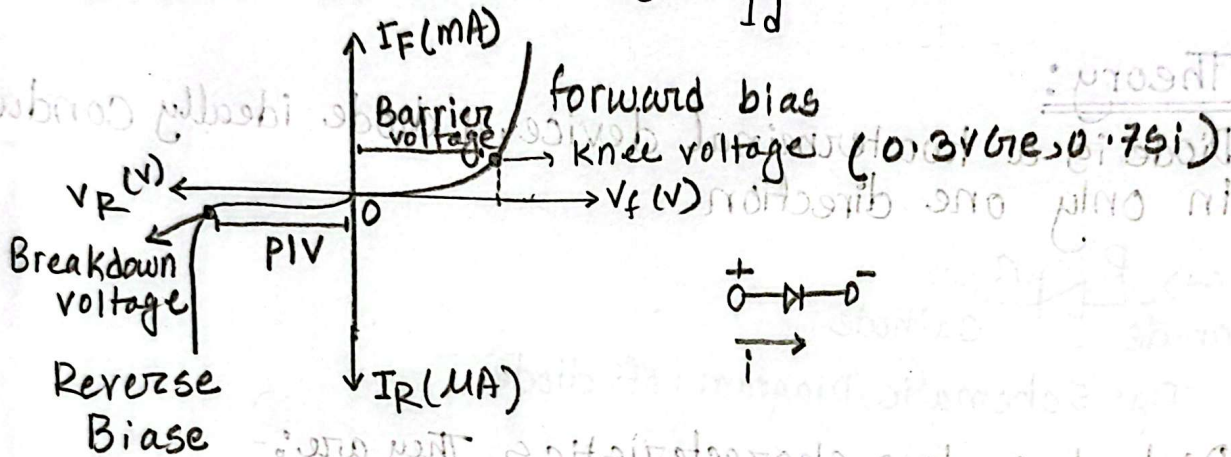


Fig: I-V characteristics of diode

List of Equipments:

Serial no	Component Details	Specification	Quantity
1.	p-n junction diode	1N4007	1 piece
2.	Resistor	1 k Ω	1 piece
3.	DC power supply		1 unit
4.	Digital Multimeter		1 unit
5.	Chords and wire		as required

Circuit Diagram:

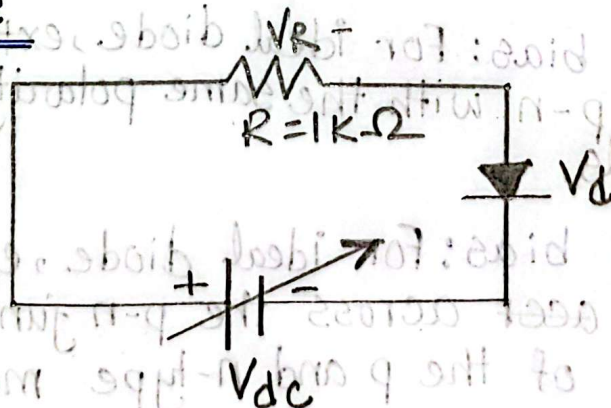


Fig: Diode forward biased configuration

Data Table: $R_{\text{theoretical}} = 1 \text{ k}\Omega$; $R_{\text{measured}} = 0.994 \text{ k}\Omega$

$V_{dc} \text{ (Volt)}$	$V_d \text{ (Volt)}$	$V_R \text{ (Volt)}$	$I_d = V_R / R \text{ (mA)}$
0.1	0.07	0.00	0
0.3	0.33	0.00	0
0.5	0.46	0.04	0.04
0.7	0.51	0.16	0.16
0.9	0.55	0.36	0.36
1	0.56	0.47	0.47
2	0.62	1.40	1.41
3	0.64	2.39	2.40
4	0.65	3.35	3.37
6	0.68	5.37	5.40
8	0.69	7.32	7.36
10	0.70	9.26	9.32
12	0.71	11.23	11.30
14	0.72	13.26	13.34
16	0.72	15.19	15.28
18	0.73	17.27	17.37
20	0.74	19.31	19.43

2 Row calculation, $V_R = 0.62$, $V_R = 0.3.35$
 So, $I_d = \frac{0.62 \times 3.35}{0.994} = 3.37 \text{ mA}$

$V_R = 19.31$, so $I_d = \frac{19.31}{0.994} = 19.43 \text{ mA}$

Result: In the diagram, we see a series circuit and by applying KVL, we get, $V_{dc} = V_d + V_R$. From the collected data we observed that, we almost got the accurate result. As the circuit is a series circuit so $I = I_d = I_R$. The minor fluctuations occurred due to temperature and instrument precision which is multimeter sensitivity.

Question Answer:

What is dynamic and static resistance of a diode?

Ans: The resistance offered by the diode when AC is applied to the diode is called AC resistance or dynamic resistance. The current flows in both the direction when AC voltage is applied. The ratio of change in voltage to change in current represent dynamic resistance of the diode. It is denoted by r_{ac} .

$$r_{ac} = \frac{\text{Change in voltage}}{\text{Change in Current}} ; r_{ac} = \frac{\Delta V}{\Delta I}$$

When DC is fed to diode, the current flows in one direction. This resistance offered by the diode is called the DC resistance.

From the graph, find V_d for corresponding values of $I_d = 5\text{mA}$ and $I_d = 10\text{mA}$ and calculate the static resistance.

From graph,

$$I_d = 5\text{mA}$$

$$V_d = 0.68\text{V}$$

$$\therefore R = \frac{V_d}{I_d}$$

$$= \frac{0.68\text{V}}{5\text{mA}}$$

$$= 0.136\text{ K}\Omega$$

$$= 136\Omega$$

$$I_d = 10\text{mA}$$

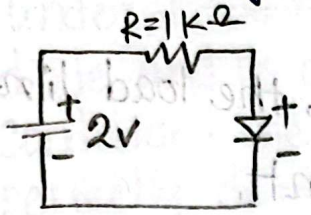
$$V_d = 0.70\text{V}$$

$$\therefore R = \frac{0.70\text{V}}{10\text{mA}}$$

$$= 0.07\text{ K}\Omega$$

$$= 70\Omega$$

Considering $V_{dc} = 2$ volt, find the load line.



here, $R = 1\text{ k}\Omega$

$V_{dc} = 2\text{ V}$

As series circuit, $I_D = I_R$

When, load line $V_d = 0$

$$KVL, V_{dc} = V_R + V_d = V_R + 0$$

$$\Rightarrow V_{dc} = I_D R_L \quad [\text{as } V = IR]$$

$$\therefore I_D = \frac{V_{dc}}{R_L} = \frac{2\text{ V}}{1\text{ k}\Omega} = 2\text{ mA}$$

When $I_D = 0$,

$$V_{dc} = V_R + V_d$$

$$\Rightarrow V_{dc} = I_D R_L + V_d$$

$$\Rightarrow V_{dc} = 0 \times R_L + V_d \quad \therefore V_{dc} = V_d = 2\text{ V}$$

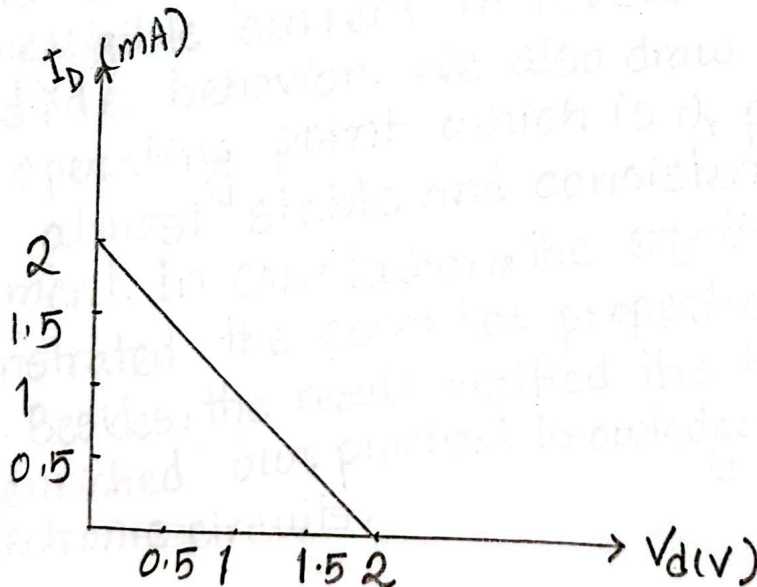


Fig: Graphical representation of load line

Draw the load line in the I_d vs V_d curve of diode and find Q-point.

From the graph we see that, the load line intercept in $(0.62, 1.3)$ point.

$$\therefore V_d = 0.62\text{V} \text{ and } I_d = 1.3\text{mA}$$

This intercepting point is Q point.

Therefore, The point Q is $(0.62, 1.3)$.

$$\text{When } I_D = 0,$$

$$V_{DC} = V_R + V_D$$

$$\Rightarrow V_{DC} = I_D R_L + V_D$$

$$\Rightarrow V_{DC} = 0 \times R_L + V_D$$

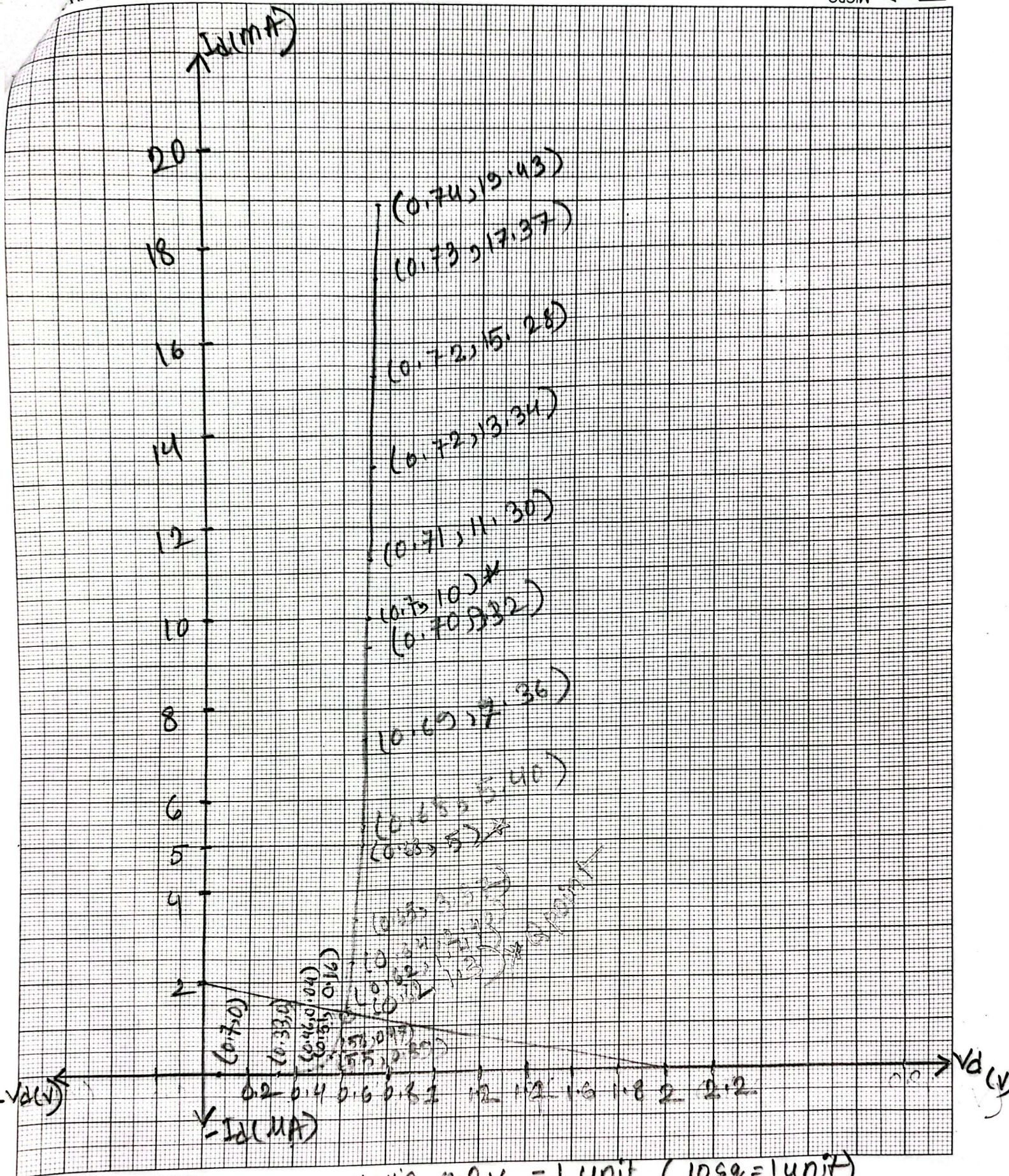
$$V_{DC} = 6\text{V} = 6\text{V} \therefore$$

Discussion: Through this experiment, we gained a clear understanding of the practical behavior of a p-n junction diode and its current-voltage relationship under different bias conditions. The I-V characteristic obtained from the experiment perfectly demonstrates the fundamental property of a diode - that it conducts current only in the forward direction after reaching a certain threshold voltage. Initially, as the forward voltage increased, the current remained almost zero until around 0.7 V, beyond which the current rose sharply, indicating the diode's cut-in voltage. This behavior closely aligns with the theoretical expectation for a silicon diode, confirming that the device allows current flow only after overcoming the potential barrier of approximately 0.7 V. The overall experiment results show excellent agreement with theoretical predictions. The plotted I-V curve exhibits the expected exponential rise in forward bias and negligible current in reverse bias, reflecting the ideal diode behavior. We also draw load line and got our operating point which is Q point. The readings were almost stable and consistent throughout the experiment. In conclusion, the experiment successfully demonstrated the essential properties of a semiconductor diode. Besides, the result verified the theory with precision and enriched our practical knowledge of diode operation in electronic circuits.

I_d vs V_d curve

Q1+Q5+Q2

MICRO



X-axis 0.2 V = 1 unit (10sq = 1 unit)
Y-axis 2ma = 1 unit (20sq = 1 unit)

Roll No.

